Wagering Theory Research Log

Jeopardy! Wagering Strategy — Research Foundations

Status: working notes. Consolidated into the finalized, citable version at `docs/white-paper.md` on 2026-07-23 — read that one for the polished version of everything below, including worked examples and the current validation status. This file stays as the running research log. Date: 2026-07-23

1. Scope

Build a wagering strategy model for two decisions: Daily Double (DD) bets and Final Jeopardy (FJ) bets. The primary case is a 3-player game. 2-player and 1-player games are reduced cases of the same model.

2. Data landscape

J-Archive (j-archive.com) is the community record of historical games. Its own site terms forbid scraping, republication, and monetization of its content. A scraper built against J-Archive directly would break those terms.

Three usable alternatives exist:

- **jwolle1/jeopardy_clue_dataset** (GitHub). 538,845 clues through Season 41. Each row lists the clue value, the Daily Double wager (zero if not a DD), round, and category. A companion file gives each contestant's score after the Single, Double, and Final rounds. The README states two restrictions: the source "prefers not to be credited," and the data must not be used "to make a public-facing web site, app, or any other product." This makes the dataset usable for internal calibration and validation only — not for a shipped product, white paper appendix, or public dataset release.
- **J-Archive's own Wagering Calculator** (j-archive.com/wageringcalculator.php). A reference implementation of community FJ logic. Useful to compare our model's output against, not a data source.
- **Tesauro, Gondek, Lenchner, Fan & Prager (2013), "Analysis of Watson's Strategies for Playing Jeopardy!," Journal of Artificial Intelligence Research 47:205–251** (arXiv:1402.0571). The IBM Watson team mined roughly 3,000 J-Archive episodes to build empirical models of DD accuracy, FJ wagering behavior, and DD placement. The paper publishes the resulting aggregate statistics. These numbers are citable research findings, not raw scraped rows, so they carry no reuse restriction.

Working decision: cite the Tesauro et al. published statistics for anything that ships (white paper, deck, docs). Use the jwolle1 dataset only to validate our simulator internally, never to populate a shipped artifact. See ADR-0001.

3. Final Jeopardy wagering

3.1 Two-player heuristics (Keith Williams, "The Final Wager")

- **Lockout:** if the leader (A) holds more than double the second-place score (B), A cannot lose regardless of the FJ outcome and does not need to wager.

- **Cover bet:** if A holds less than double B, A should wager at least `2B - A` to stay ahead even if B doubles up and A misses.
- **Bet-to-tie:** in some score ranges, wagering to land on an exact tie beats a full cover bet, because it changes which outcomes A needs to win.
- **Two-thirds rule** (Dupee, 1998): if B holds less than two-thirds of A's score, B's expected value from wagering zero (hoping A misses) can exceed the expected value of trying to overtake A. This is a heuristic threshold. The true break-even point depends on each player's real confidence in the category, which opponents cannot observe.

3.2 Three-player heuristics

Williams frames the three-player case as three pairwise sub-problems: A vs. B, B vs. C, and A vs. C. The wager must satisfy all three constraints at once, which sometimes forces a tie-wager instead of a clean cover.

Tesauro et al. found that human C-place (third-place) wagers show almost no dependence on game state — third-place bettors behave close to randomly. A and B, by contrast, show strong, learnable patterns; the Watson team split their behavior into six empirically distinct regimes based on B's score relative to A and to C.

3.3 Game-theoretic layer

The exact optimum requires each player's private confidence in the FJ category, which opponents cannot observe — formally, a Bayesian game. The Watson team avoided computing a full Bayes-Nash equilibrium (too costly to compute, and human opponents do not play equilibrium strategy anyway). Instead they computed a **Best-Response** strategy against an empirically fitted model of human wagering, using Monte Carlo simulation.

Implication for this project: build two layers.

1. An interpretable heuristic layer (Williams' addition/subtraction rules, the two-thirds rule, lockout/lock-tie detection) for at-a-glance use.
2. A simulation-backed optimizer (Monte Carlo best-response against a fitted opponent model) for practice, analysis, and to validate layer 1.

Layer 2 is a study and practice tool, not a live-assist device — live wagering assistance during an actual taping would break the show's rules. State this explicitly in the README and white paper.

Notable edge cases the model must handle: lockout, lock-tie (B holds exactly half of A's score — B should bet everything, since a tie beats needing both a correct answer and an A miss), and "anti-two-thirds" bets in categories where confidence is unusually low.

4. Daily Double wagering

4.1 Empirical baseline

Tesauro et al. report DD accuracy varies strongly by round and row (a proxy for clue difficulty): roughly 72% on the easiest (top) row of second-round DDs, falling to roughly 57% on the hardest (bottom) row. A full breakdown by round and row needs a closer read of the paper's tables before it goes into our model's calibration data.

The paper's central finding on human behavior: contestants underbet DDs relative to the wager that maximizes win probability. In last-DD situations (second round, no DDs remaining), leading players bet $2,590 on average against a $5,380 recommended optimum; trailing players bet $3,120 against a recommended $6,970. Quantifying this conservatism gap — how much win probability a real bet leaves on the table — is a natural benchmark output for our tool.

4.2 Proposed expected-value model

State: current score, opponent scores, remaining board value, round, and a confidence estimate `p` for the category (row position is one proxy; self-reported confidence is another).

```
E(bet) = p · V(score + bet, ...) + (1 - p) · V(score - bet, ...)
```

`V(...)` is the win probability of the resulting state, estimated by simulating the rest of the game forward (Monte Carlo rollout). Search should include near-miss bets around "lock-tie" and "lockout" score targets — the Watson paper shows these off-the-obvious-number bets can carry outsized value because of discrete tie outcomes.

4.3 Generalizing to three (and fewer) players

With three players, `V(...)` must track standing against two opponents at once — every DD or FJ decision branches into beat-both, beat-one, or beat-none outcomes. This three-way branching is the central technical challenge; most public tools, including the pairwise decomposition in Williams' own three-player tutorial, approximate it rather than solving it jointly. Two-player and one-player modes are degenerate cases of the same state representation, so the engine should be built once for N players and specialized for N = 1, 2, 3.

5. Proposed architecture (draft)

- **Language:** Python.
- **Core:** an N-player game-state simulator plus a Monte Carlo win-probability estimator, calibrated on per-round/per-row accuracy rates and empirical opponent-wagering distributions.
- **Two consumer layers:** the closed-form heuristic module (section 3.1, 3.2 rules) and the simulation-backed optimizer (section 3.3, 4.2), with the optimizer used to validate the heuristics and to produce the white paper's quantitative claims.
- **Validation:** reproduce the Tesauro et al. headline numbers (DD accuracy by row, human wager conservatism, two-thirds threshold) as a calibration check before trusting the model's three-player extensions.

6. Open questions and risks

See `RISK_REGISTER.md` at the project root for the tracked version. Draft entries:

1. Data licensing limits what can appear in shipped artifacts (see section 2).
2. Real confidence is unobservable in historical data; the tool likely needs a self-reported confidence input rather than a purely inferred one.
3. Live in-game assistance would break contest rules; scope this as a study and post-game analysis tool only.
4. Pairwise heuristic decomposition for three players is an approximation, not a proven joint optimum; needs empirical validation against the full simulator.

7. Status (updated 2026-07-23)

Done:

1. Scope confirmed: FJ first, since its guarantee-based math needs no confidence/probability inputs ("more fixed"). Data-source approach (ADR-0001) accepted.
2. `engine/fj_wager.py` (Tier 1, guarantee-based FJ) built and tested.
3. `engine/fj_bestresponse.py` (Tier 2, confidence- and opponent-model-based FJ) built and tested. See ADR-0002 for the opponent-model assumptions.
4. `engine/simulate.py` and `engine/dd_wager.py` (Daily Double, Monte Carlo) built and tested.
5. 26 tests passing across all four modules.

Still open:

1. Full project scaffold (README, LICENSE, SECURITY.md, CONTRIBUTING.md, CI) – not started; CI specifically needs a GitHub remote, which isn't connected yet.
2. Calibrate the Tier 2 / DD engines' placeholder constants against Tesauro et al.'s full published tables (ADR-0002, risk #5).
3. Write the white paper, deck, and Diátaxis technical docs once the engines are calibrated and validated, not just directionally correct.
4. Git: one local commit exists; further commits are blocked by a sandbox quirk (see CHANGELOG 0.2.0 notes) until Will clears it on his end.

Future ideas (not built, just logged):

1. Surface "mathematically eliminated" rivals explicitly in the app UI – e.g., "Rival 2's score doesn't affect this wager." The underlying math in `fj_wager.py` already ignores such rivals correctly (it only ever looks at the single highest opponent ceiling, never a sum or average across all opponents) – see white paper section 4.4 for the full explanation. This would be a usability enhancement, not a math fix.

Sources

- Tesauro, G., Gondek, D., Lenchner, J., Fan, J., & Prager, J. M. (2013). Analysis of Watson's strategies for playing Jeopardy!. Journal of Artificial Intelligence Research, 47, 205–251. https://arxiv.org/pdf/1402.0571
- Williams, K. "How to wager in Final Jeopardy!" The Final Wager. https://thefinalwager.wordpress.com/tutorial/
- jwolle1. jeopardy_clue_dataset. GitHub. https://github.com/jwolle1/jeopardy_clue_dataset
- J! Archive Wagering Calculator. https://j-archive.com/wageringcalculator.php